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Skill week – 3

Step 1 – Create Database, Tables and Insert Class Data

The screenshot shows the MySQL Workbench interface. The 'Query 1' window contains the following SQL code:

```
1 CREATE DATABASE SkillWeek3;
2
3 USE SkillWeek3;
4
5 CREATE TABLE class (
6     id INT,
7     name VARCHAR(30)
8 );
9
10 CREATE TABLE class_info (
11     id INT,
12     address VARCHAR(30)
13 );
14
15 INSERT INTO class VALUES
16 (1, 'abhi'),
17 (2, 'adam'),
18 (4, 'alex');
```

The 'Output' window shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	19:20:51	CREATE DATABASE SkillWeek3	1 row(s) affected	0.000 sec
2	19:20:51	USE SkillWeek3	0 row(s) affected	0.000 sec
3	19:20:51	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected	0.032 sec
4	19:20:51	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected	0.015 sec
5	19:20:51	INSERT INTO class VALUES (1, 'abhi'), (2, 'adam'), (4, 'alex')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec

Step 2 – CROSS JOIN

The screenshot shows the MySQL Workbench interface. The 'Query 1' window contains the following SQL code:

```
1 INSERT INTO class_info VALUES
2 (1, 'DELHI'),
3 (2, 'MUMBAI'),
4 (3, 'CHENNAI');
```

The 'Output' window shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	19:21:18	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec

The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL code:

```
1 SELECT *
2 FROM class
3 CROSS JOIN class_info;
```

The result grid displays the following data:

	id	name	id	address
4	alex	1	DELHI	
2	adam	1	DELHI	
1	abhi	1	DELHI	
4	alex	2	MUMBAI	
2	adam	2	MUMBAI	
1	abhi	2	MUMBAI	
4	alex	3	CHENNAI	
2	adam	3	CHENNAI	

The bottom status bar shows the message: "1 19:21:51 SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000" and "9 row(s) returned".

Step 3 – INNER JOIN

The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL code:

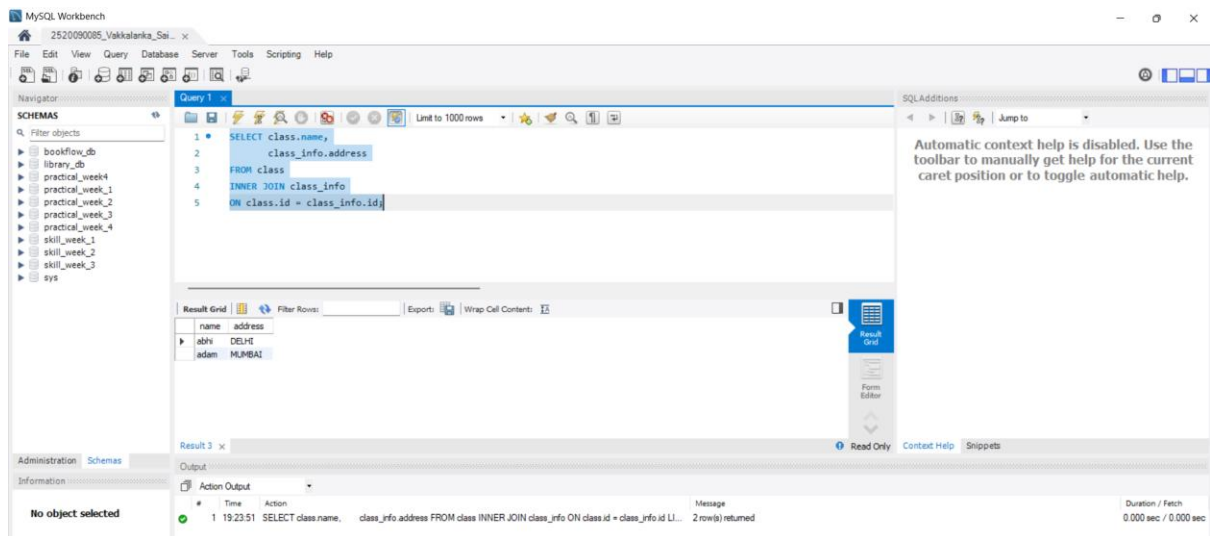
```
1 SELECT *
2 FROM class
3 INNER JOIN class_info
4 ON class.id = class_info.id;
```

The result grid displays the following data:

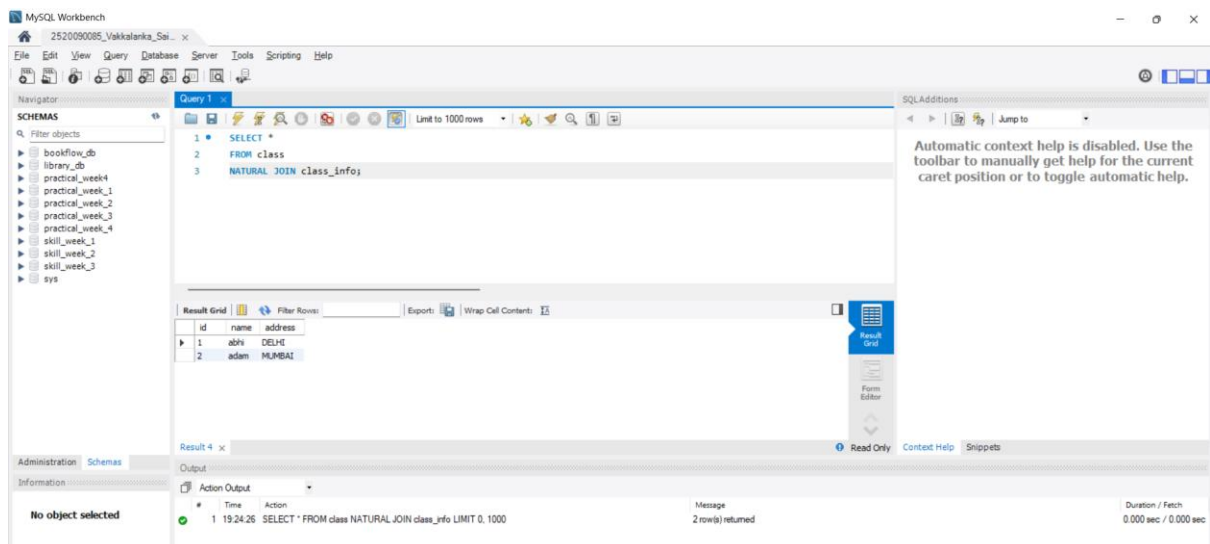
	id	name	id	address
1	abhi	1	DELHI	
2	adam	2	MUMBAI	

The bottom status bar shows the message: "1 19:23:00 SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000" and "2 row(s) returned".

Step 4 – INNER JOIN with Selected Columns



Step 5 – NATURAL JOIN



Step 6 – LEFT OUTER JOIN

MySQL Workbench

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File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bookflow_db
- library_db
- practical_week4
- practical_week_1
- practical_week_2
- practical_week_3
- practical_week_4
- skill_week_1
- skill_week_2
- skill_week_3
- sys

Query 1

```

1 SELECT *
2 FROM class
3 LEFT OUTER JOIN class_info
4 ON class.id = class_info.id;

```

Result Grid

	id	name	id	address
1	abhi	1	DELHI	
2	adam	2	MUMBAI	
4	alex			
5	ashish			

Result 5

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	19:26:45	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

No object selected

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Step 7 – Find Unmatched Rows from the Left Table

MySQL Workbench

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File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bookflow_db
- library_db
- practical_week4
- practical_week_1
- practical_week_2
- practical_week_3
- practical_week_4
- skill_week_1
- skill_week_2
- skill_week_3
- sys

Query 1

```

1 SELECT *
2 FROM class
3 LEFT JOIN class_info
4 ON class.id = class_info.id
5 WHERE class_info.id IS NULL;

```

Result Grid

	id	name	id	address
4	alex			
5	ashish			

Result 6

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	19:27:26	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT 0, 2 row(s) returned	2 row(s) returned	0.000 sec / 0.000 sec

No object selected

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Step 8 – RIGHT OUTER JOIN

MySQL Workbench

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File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bookflow_db
- library_db
- practical_week4
- practical_week_1
- practical_week_2
- practical_week_3
- practical_week_4
- skill_week_1
- skill_week_2
- skill_week_3
- sys

Query 1

```

1 SELECT *
2 FROM class
3 RIGHT OUTER JOIN class_info
4 ON class.id = class_info.id;

```

Result Grid

	id	name	id	address
1	abhi	1	DELHI	
2	adam	2	MUMBAI	
3	alex	3	CHENNAI	
7		7	NOIDA	
8		8	PANIPAT	

Result 7

Output

Action Output

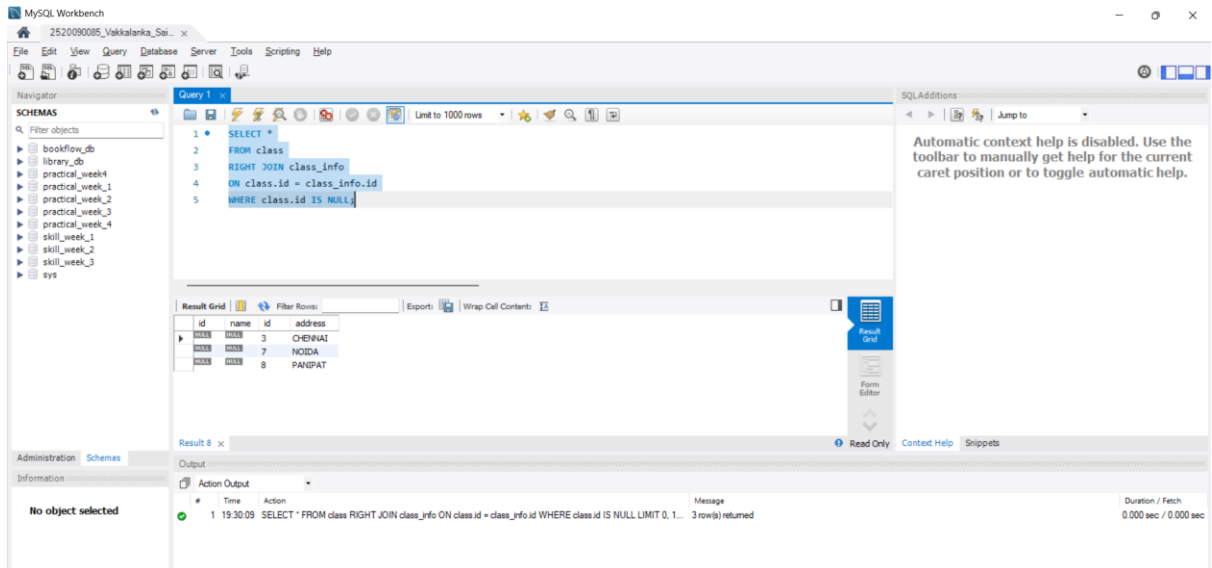
#	Time	Action	Message	Duration / Fetch
1	19:29:21	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

No object selected

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Step 9 – Find Unmatched Rows from the Right Table



The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

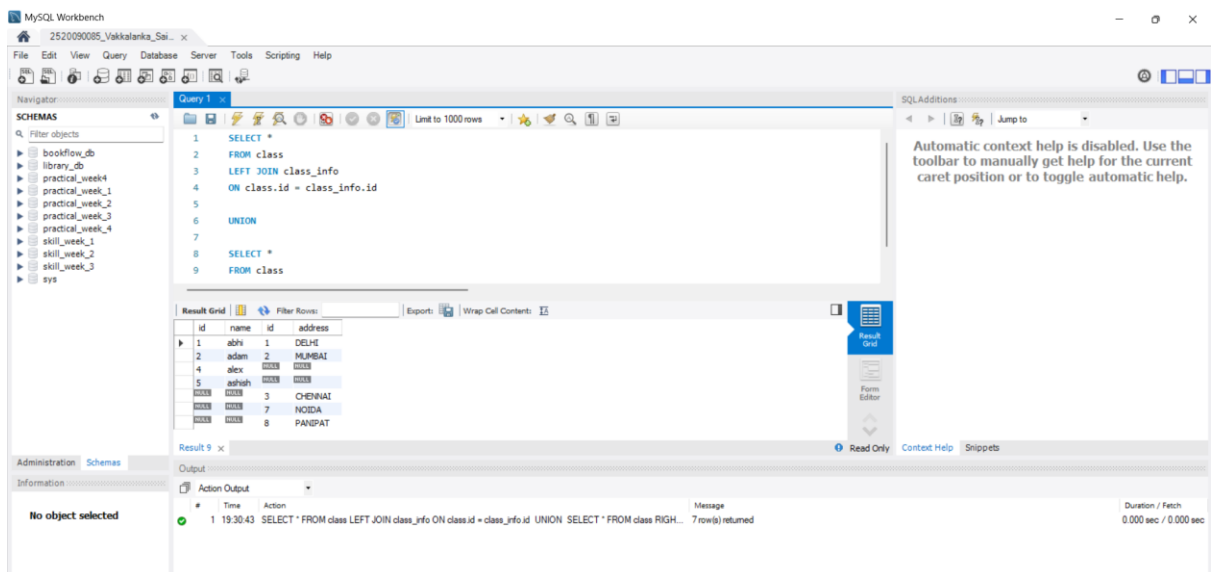
```
1 SELECT *
2 FROM class
3 RIGHT JOIN class_info
4 ON class.id = class_info.id
5 WHERE class.id IS NULL
```

The result grid displays the following data:

id	name	id	address
NULL	NULL	3	CHENNAI
NULL	NULL	7	NOIDA
NULL	NULL	8	PANPAT

The bottom status bar shows the message: "1 19:30:09 SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class.id IS NULL LIMIT 0, 1... 3 row(s) returned".

Step 10 – Simulate FULL OUTER JOIN



The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

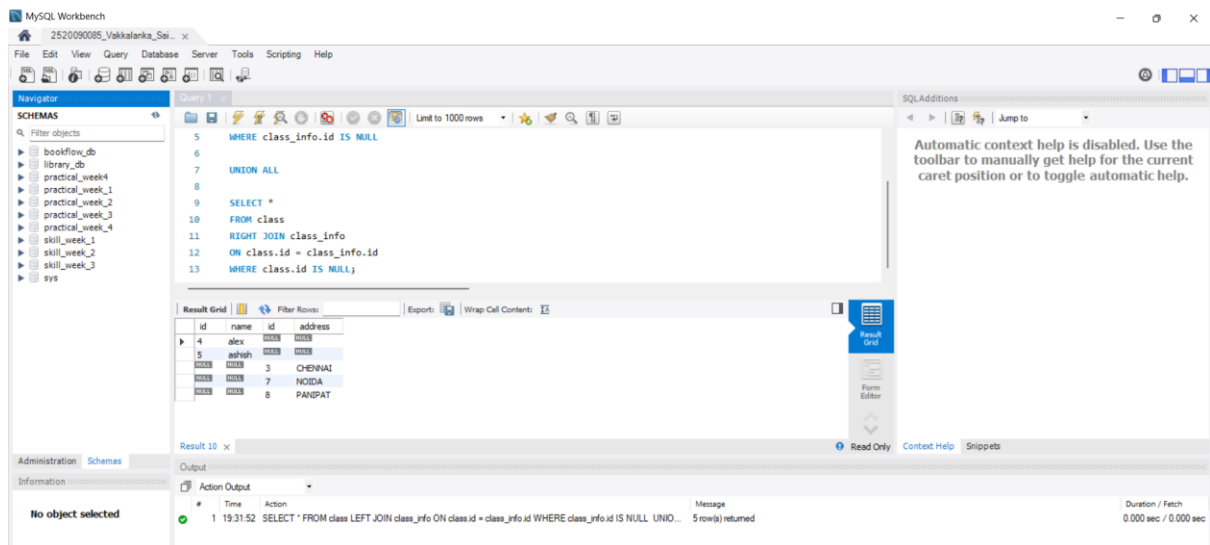
```
1 SELECT *
2 FROM class
3 LEFT JOIN class_info
4 ON class.id = class_info.id
5
6 UNION
7
8 SELECT *
9 FROM class
```

The result grid displays the following data:

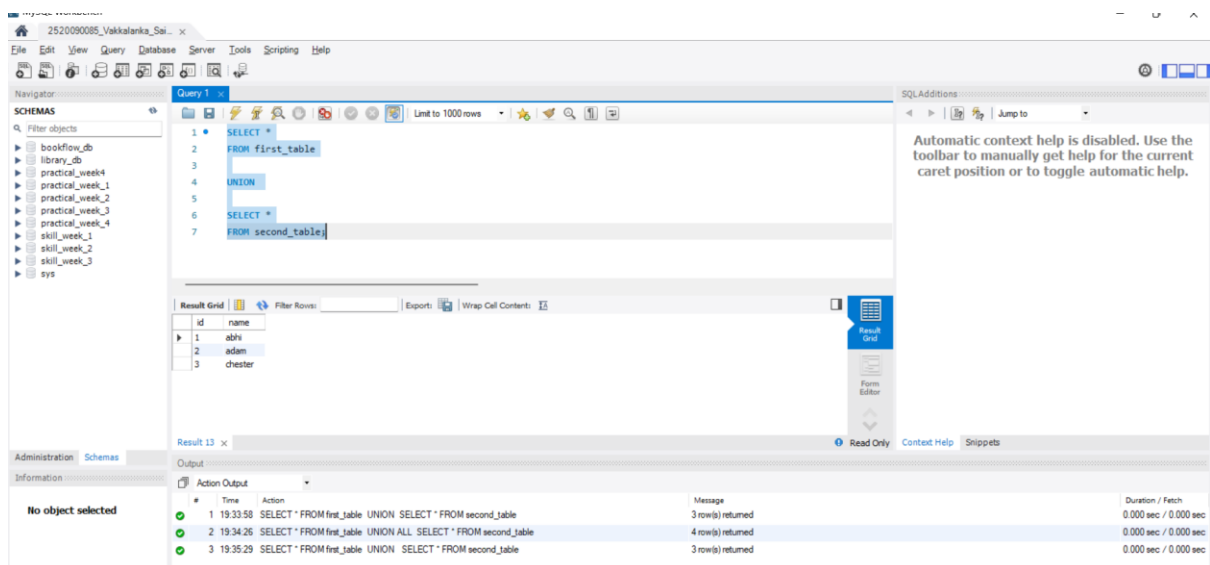
id	name	id	address
1	abhi	1	DELHI
2	adnan	2	MUMBAI
4	alex	NULL	NULL
5	ashish	NULL	NULL
NULL	NULL	3	CHENNAI
NULL	NULL	7	NOIDA
NULL	NULL	8	PANPAT

The bottom status bar shows the message: "1 19:30:43 SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id UNION SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class.id IS NULL LIMIT 0, 1... 7 row(s) returned".

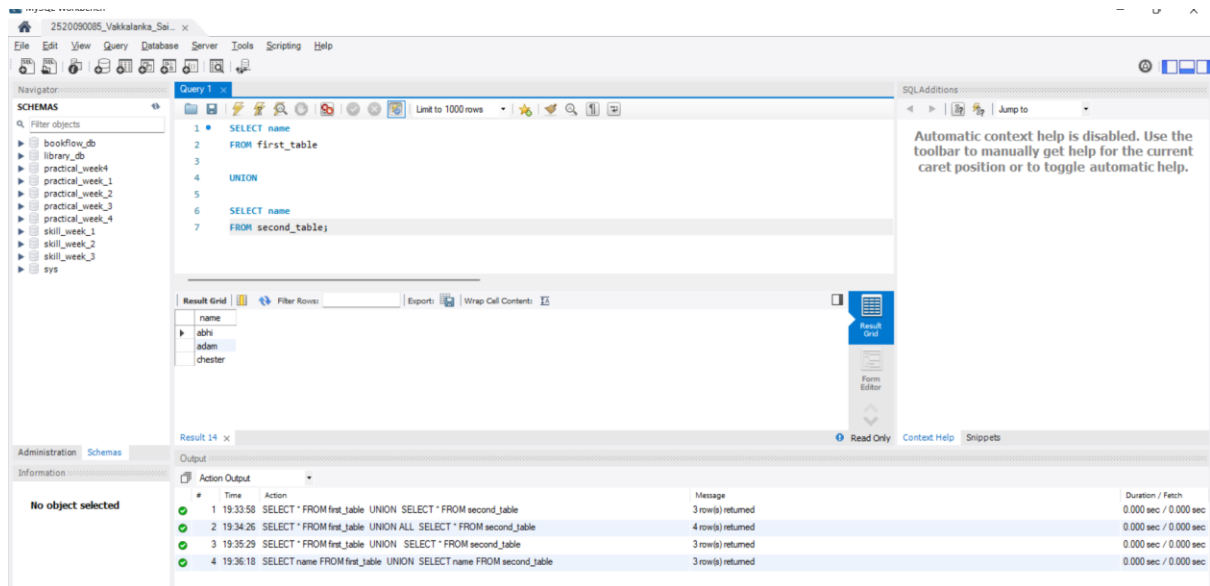
Step 11 – FULL OUTER JOIN for Unmatched Rows



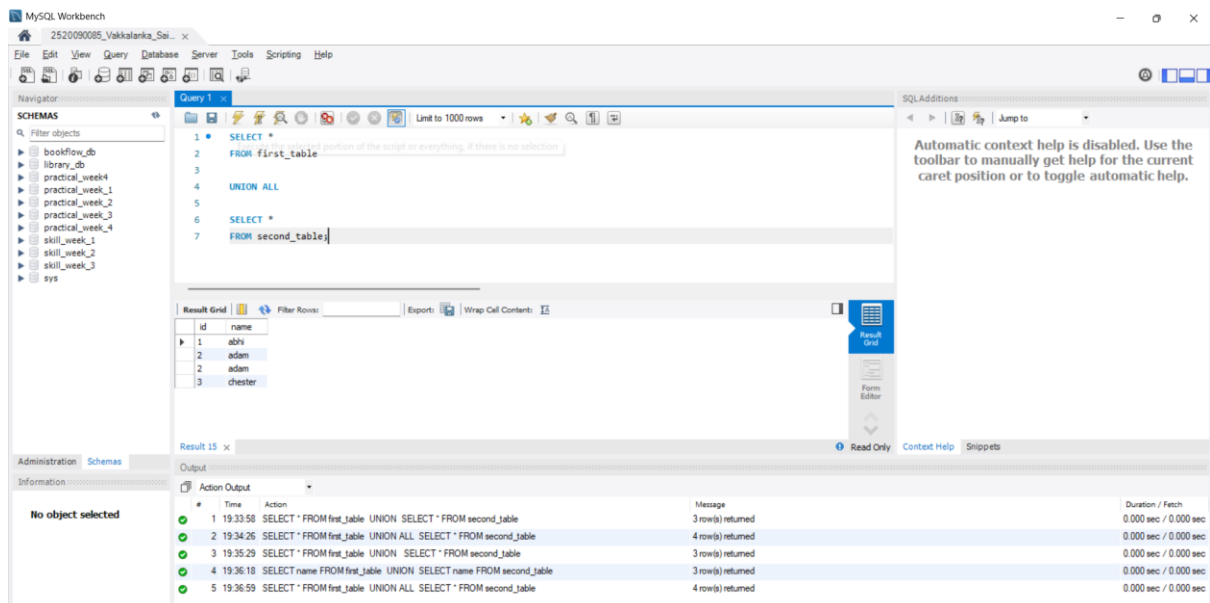
Step 12 – UNION of Two Tables



Step 13 – UNION on a Single Column



Step 14 – UNION ALL



Step 15 – COUNT with UNION ALL

The screenshot shows the MySQL Workbench interface. The 'Query 1' editor contains the following SQL code:

```
3 (
4   SELECT *
5   FROM first_table
6
7   UNION ALL
8
9   SELECT *
10  FROM second_table
11 ) AS Adj
```

The 'Result Grid' shows a single row with the value 4 under the column COUNT(*).

The 'Action Output' pane shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
1	19:33:58	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
2	19:34:26	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned	0.000 sec / 0.000 sec
3	19:35:29	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
4	19:36:18	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
5	19:36:59	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned	0.000 sec / 0.000 sec
6	19:37:24	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table)	1 row(s) returned	0.016 sec / 0.000 sec

Step 16 – INNER JOIN Using Table Aliases

The screenshot shows the MySQL Workbench interface. The 'Query 1' editor contains the following SQL code:

```
1 SELECT c.id,
2        c.name,
3        ci.address
4 FROM class c
5 INNER JOIN class_info ci
6 ON c.id = ci.id
```

The 'Result Grid' shows the following data:

id	name	address
1	abhi	DELHI
2	adnan	MUMBAI

The 'Action Output' pane shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
1	19:33:58	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
2	19:34:26	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned	0.000 sec / 0.000 sec
3	19:35:29	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
4	19:36:18	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
5	19:36:59	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned	0.000 sec / 0.000 sec
6	19:37:24	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table)	1 row(s) returned	0.016 sec / 0.000 sec
7	19:38:10	SELECT c.id, c.name, ci.address FROM class c INNER JOIN class_info ci ON c.id = ci.id LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

Step 17 – CASE Expression with LEFT JOIN

MySQL Workbench

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File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

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- practical_week4
- practical_week1
- practical_week2
- practical_week3
- practical_week4
- skill_week1
- skill_week2
- skill_week3
- sys

Query 1: x

Limit to 1000 rows

```
2 c.name,  
3 CASE  
4 WHEN ci.address IS NULL  
5 THEN 'Address Missing'  
6 ELSE 'Address Available'  
7 END AS Status  
8 FROM class c  
9 LEFT JOIN class_info ci  
10 ON c.id = ci.id;
```

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

id	name	Status	desc
1	abhi	Address	describe
2	adani	Address	do
4	alex	Address	drop
5	ashish	Address	missing

Administration Schemas

Information

No object selected

Output

#	Time	Action	Message	Duration / Fetch
2	19:34:26	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned	0.000 sec / 0.000 sec
3	19:35:29	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
4	19:36:18	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
5	19:36:59	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned	0.000 sec / 0.000 sec
6	19:37:24	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_...	1 row(s) returned	0.016 sec / 0.000 sec
7	19:38:10	SELECT c.id, c.name, ci.address FROM class c INNER JOIN class_info ci ON c.id = ci.id LIMIT 0, 10...	2 row(s) returned	0.000 sec / 0.000 sec
8	19:38:56	SELECT c.id, c.name, CASE WHEN ci.address IS NULL THEN 'Address Missing' ...	4 row(s) returned	0.000 sec / 0.000 sec